

Day-14

Port no. → use to identify service

DNS - Domain Name System

It is a service used to provide name to IP address.

It is working with UDP protocol.

Port number of DNS = 53.

* If many servers are there in data centre then it is difficult to know all the IP. So engineer will create a database & give it to DNS.

PORT NUMBERS: (X)

DNS:

DNS - 53

Http - 80

https - 443

RDP - 3389

Telnet - 23

SSH - 22

NTP - 123

AD: 389

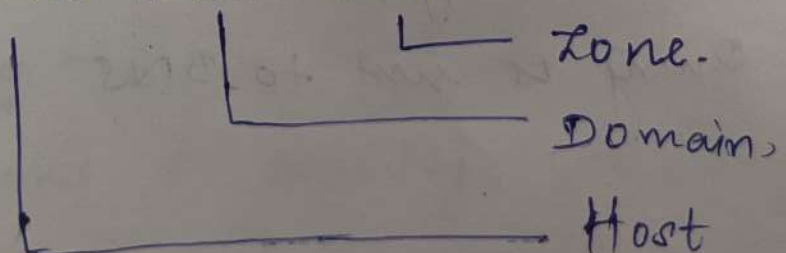
Hotstar - 172.16.0.10

Netflix - 172.16.0.20

→ We can type hotstar from client server then the webpage will come.

FQDN - Full Qualified Domain Name

WWW.HOTSTAR.COM



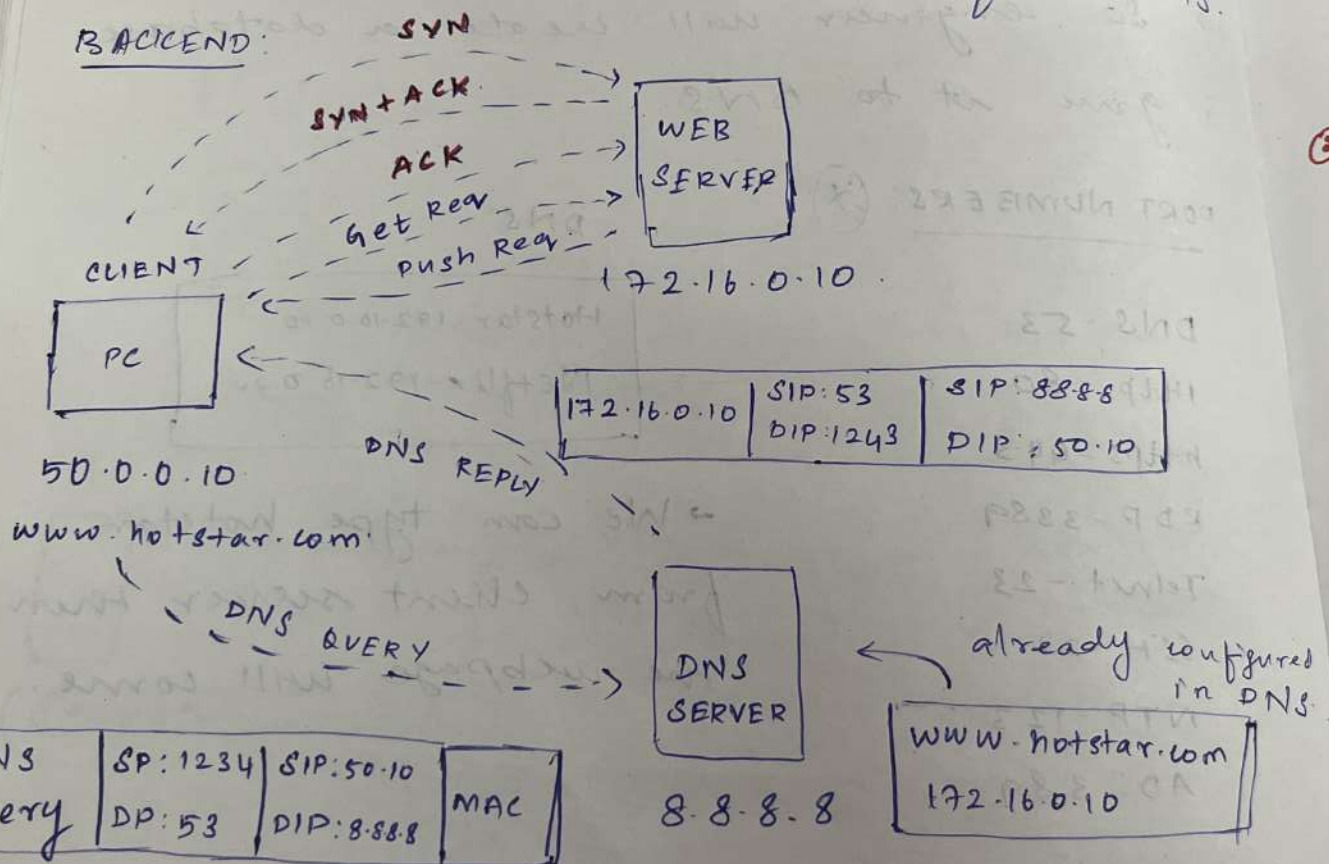
There are 2 thing in DNS:

Forward lookup zone → Giving name then IP will come

Reverse lookup zone → Giving IP to name

- We have to open DNS → Configure IP
- Install DNS → Give zone, domain & host
- open client → ncpa-cpl → preferred DNS.

BACKEND:



→ When user types the `www.hotstar.com`, DNS Query is sent to DNS.

Source port: Random (1234)

Destination port: DNS (53)

Source IP & Dest. IP

MAC encapsulation

} → Layer 4 process.

→ Now DNS gives reply with IP of the hoststar. we know the IP now.

① SYN →

SYN	SP: 1234 DP: 80	SIP: 50.10 DIP: 192.16.0.10	MAC
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DP: 80 (http):

② SYN + ACK →

SYN/ACK	SP: 80 DP: 1234	SIP: 192.16.0.10 DIP: 50.10	MAC
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③ ACK →

ACK	SP: 1234 DP: 80	SIP: 50.10 DIP: 192.16.0.10	MAC
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→ After 3-way handshake:

1) Get Req.

2) Push Req.

When client wants to access a web page using name → DNS Query (asking IP of web page)

→ DNS Response (gives IP) → Now client PC knows the IP so it uses TCP protocol → SYN → SYN + ACK

→ ACK → Get Req → Push Req.

When client uses IP directly, DNS Query & DNS Response will not be there.